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Ultrasonic-assisted wire arc additive manufacturing significantly improves impact toughness of 5183 aluminum alloy

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ABSTRACT

Wire arc additive manufacturing (WAAM) is a promising method for fabricating large-scale, lightweight aluminum alloy structures due to its high deposition rate, material efficiency, and design flexibility. However, as-deposited WAAM aluminum alloys often suffer from porosity and predominantly columnar grain structures, both of which compromise mechanical properties and hinder industrial adoption. In this study, an innovative ultrasonic-assisted WAAM (UA-WAAM) approach is proposed. A comparative analysis of porosity, microstructure, and mechanical properties between conventional WAAM and UA-WAAM samples is conducted to systematically investigate the effects of ultrasonic vibration. Microstructural analysis confirms significant grain refinement and reduced residual stress in UA-WAAM samples, attributed to ultrasonic cavitation disrupting epitaxial grain growth. Furthermore, the introduced ultrasonic vibration effectively suppresses porosity by enhancing molten pool dynamics and reducing crack initiation sites. Consequently, the UA-WAAM material exhibits a 76% higher impact toughness than conventional WAAM. Elongation increases by 49% in the vertical direction and 21% in the horizontal direction, while the ultimate tensile strength is improved by 12% in both orientations.

1. Introduction

Aluminum alloys, particularly the 5xxx and 6xxx series widely used in shipbuilding, offer excellent weldability, low density, and high corrosion resistance in marine environments. These properties enable weight reductions of up to 50–67% compared to steel structures while maintaining structural performance, though considerations such as localized corrosion and fire safety must be addressed. Indeed, AA5183 aluminum alloy is extensively utilized for structural components in the shipbuilding industry due to its high corrosion resistance and good weldability [1]. Conventional aluminum alloy manufacturing techniques (primarily casting and forging) suffer from long production cycles, low efficiency, and high costs. Moreover, these traditional methods struggle to meet the increasingly demanding requirements of modern industrial applications [2,3]. Additive manufacturing (AM) offers high efficiency, flexibility, and cost-effectiveness by building structures layer-by-layer, in contrast to traditional subtractive methods. This paradigm provides significant improvements in manufacturing efficiency and reduced material waste [3]. AM has rapidly emerged as a pivotal technology in aerospace, automotive, marine, and other industries. Metal AM techniques are generally categorized by heat source: laser-based (e.g. SLM), electron beam (EBAM), and arc-based (WAAM). Among these, wire arc additive manufacturing (WAAM) is an arc welding-based process that deposits metal layer by layer under program control. It offers high deposition rates, high material utilization, low cost, and good equipment flexibility, and can achieve favorable mechanical properties after process optimization [4,5]. The combination of aluminum alloys with WAAM not only fulfills the requirements for

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lightweight design and corrosion resistance in marine applications, but also enables efficient, low-cost production of large-scale components. This synergistic approach demonstrates significant technical advantages and promising prospects in the shipbuilding industry [6].

Despite its advantages, WAAM has several issues, such as unstable heat input and uneven cooling [7]. These lead to defects including porosity, persistent columnar grain growth, and high residual stresses in as-deposited builds [7]. Unstable heat input causes fluctuations in the thermal profile and non-uniform cooling rates, generating high residual stresses that can result in part deformation [8]. Among WAAM defects, porosity is common and directly degrades fatigue and tensile strength by causing local strain concentrations and premature cracking [9]. Additionally, steep thermal gradients in WAAM promote continuous columnar grain structures that severely impair mechanical properties [10]. In summary, these defects compromise the mechanical performance, structural integrity, and service life of WAAM components, impeding broader adoption in marine applications [11].

To date, many strategies have been explored to improve WAAM component performance. Zhu et al. [14] introduced an ultrasonic rotation extrusion-WAAM (URE-WAAM) process that integrates ultrasonic vibration and mechanical deformation, which significantly reduced porosity and refined the grain structure with a more uniform precipitate distribution. As a result, the mechanical properties of URE-WAAM samples improved notably, although the required equipment was complex and less efficient. Zhou et al. [15] applied inter-layer hammer peening during WAAM, achieving a periodic arrangement of coarse and fine grains, but this technique greatly reduced production efficiency. Wang et al. [16] demonstrated that an ultrasonic-frequency pulsed arc can be achieved by superimposing high-frequency pulse currents onto a conventional arc welding waveform. Kumar et al. [17] investigated ultrasonic-assisted WAAM, taking full advantage of ultrasonic vibrations: the high-frequency vibration introduced into the molten pool disrupts solidification kinetics and thereby mitigates solidification cracking. Zhang et al. [18] studied ultrasonic impact treatment (UIT), which uses impact energy to induce severe plastic deformation on the solidified surface, refining grains and introducing beneficial compressive stress. Although such mechanical ultrasonication can refine the grain structure of WAAM aluminum alloys and improve their mechanical properties, the welding apparatus becomes relatively complex and some ultrasonic energy is lost before reaching the molten pool, limiting its applicability. Xu et al. [19,20] showed that ultrasonic energy can promote rapid gas effusion from the molten pool, reducing gas entrapment and porosity. The cavitation effect from ultrasonic vibration vigorously stirs the molten pool, aiding gas release and improving weld quality. Shi et al. [21,22] demonstrated that ultrasonic-assisted WAAM can refine the grain structure of aluminum alloys. High-frequency ultrasonic vibrations promote intense stirring of the molten pool, facilitating pore escape and grain refinement, thereby enhancing mechanical properties such as strength and fatigue resistance. By adjusting the ultrasonic frequency and pulsed arc parameters, porosity, microstructure, and mechanical properties can be effectively controlled [12,13].

However, the impact toughness and fracture behavior of ultrasonically assisted WAAM aluminum alloys have not been comprehensively studied. In this study, we investigate an ultrasonic-assisted TIG (gas tungsten arc) based WAAM process for 5183 aluminum alloy. High-frequency pulsed electrical signals are coupled into the welding arc to generate ultrasonic vibrations, which are directly transmitted to the molten pool, thereby improving the deposition quality. A comprehensive comparison of microstructure, porosity, and mechanical properties between conventional WAAM and ultrasonic-assisted WAAM (UA-WAAM) builds is presented, elucidating the fundamental effects of ultrasonic excitation on the microstructure and performance of WAAM-fabricated 5183 aluminum alloy.

2. Experimental procedures

The substrate used in this study is 5083 aluminum alloy, and the additive wire is 5183 aluminum alloy. The relevant components are listed in Table 1. The ultrasonic-assisted WAAM (UA-WAAM) system employed in this study is shown schematically in Fig. 1. The UA-WAAM setup integrates two primary subsystems: a conventional WAAM apparatus and an ultrasonic vibration unit. As shown in Fig. 1a, the WAAM system consists of a Fronius TT3000 JOB GTAW power source, a three-axis automated gantry, a GTAW (TIG) torch with a cold wire feeder, a water-cooling unit, a gas shielding system, and the ultrasonic excitation unit. As illustrated in Fig. 1b, the ultrasonic system is connected in parallel to the welding power supply through an isolation coupling device. This configuration superimposes high-frequency ultrasonic pulse signals onto the low-frequency GTAW current, producing a modulated arc waveform that effectively introduces ultrasonic energy into the WAAM process. As depicted in Fig. 1c, the filler wire is melted in the arc with a wire feed angle of 15°. The tungsten electrode is maintained 4 mm above the workpiece, and the wire tip is positioned 3 mm from the tungsten electrode.

Optimized process parameters for the UA-WAAM deposition are summarized in Table 2. In wire arc additive manufacturing (WAAM), a 1.2 mm diameter wire offers good flexibility during feeding, minimizing the risk of wire jamming. Compared to finer wires, it provides adequate penetration depth and deposition width. Relative to thicker wires, it operates at a lower heat input under the same current, which facilitates control over the cooling rate and melt pool width, eliminating the need for higher currents that could accelerate tungsten electrode degradation. A preheating temperature of 60 °C was selected because a lower temperature would result in an excessively narrow melt pool during the first deposition, while a higher temperature would lead to an overly wide one. Both scenarios would adversely affect the geometry of subsequent deposited layers. Therefore, 60 °C was chosen to maintain consistent dimensions across the deposited layers. For the first deposited layer, a current of 180 A was used while other parameters were held constant. The current was then reduced by 15 A per layer until reaching 120 A, followed by a final reduction to 110 A. This stepwise current reduction strategy was implemented because the first layer (in contact with the substrate) experiences rapid cooling, requiring a higher current to achieve sufficient bead width. As the number of layers increases, the cooling rate decreases, so the current is gradually lowered. Optimal surface quality was achieved at 110 A. Throughout the build, the interlayer temperature was maintained at approximately 80 °C to preserve the deposit's structural integrity. Fig. 2a shows that in the additive manufacturing deposition layer, excessively high wire feeding speed or travel speed (mismatch between travel speed and wire feeding speed) may lead to incomplete melting of the wire, resulting in its entrapment in the molten pool. In contrast, deposition layer b (shown in Fig. 2b) exhibits a widened molten pool due to excessive current and high heat input, which causes the molten pool to spread laterally and form an irregular geometry. Therefore, the selected parameters have been optimized, and Fig. 2c demonstrates the optimized additive manufacturing morphology.

WAAM walls were built using the above parameters for both conventional (no ultrasound) and ultrasonic-assisted conditions. Test specimens for mechanical evaluation were extracted from the deposited walls via wire EDM, as illustrated in Fig. 3a. The specimen dimensions for Charpy impact and tensile tests are given in Fig. 3b. Charpy impact testing was performed at room temperature using an NI300 pendulum impact tester on samples of 55 mm × 10 mm × 4 mm. Tensile tests were carried out at room temperature using a DDL100 universal testing machine at a constant crosshead speed of 1.0 mm/min. All tensile and impact tests were conducted on three samples for each condition, and the average values are reported. The impact fracture surfaces were subsequently examined by scanning electron microscopy (SEM) to assess fracture modes.

Table 1
Chemical compositions (wt%) of the 5083 aluminum alloy substrate and ER5183 filler wire.

Element	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Al
5083	0.36	0.37	0.09	0.7	4.3	0.15	0.20	0.12	Rem.
ER5183	0.25	0.35	0.10	0.8	5.0	0.14	0.09	0.14	Rem.

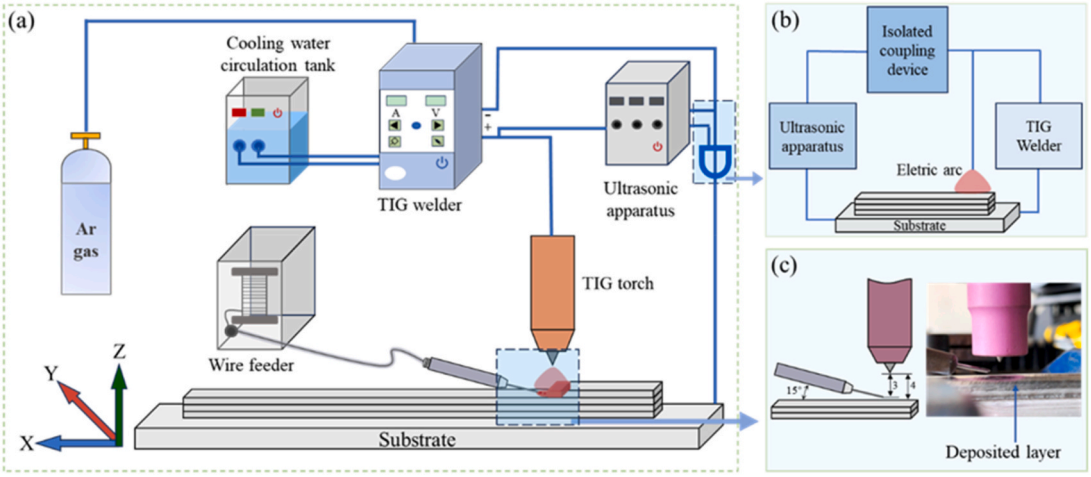


Fig. 1. Schematic diagrams of the UA-WAAM experimental system: (a) overall equipment configuration; (b) ultrasonic assistance circuitry; (c) detailed arrangement of the tungsten electrode and filler wire (relative positions).

Table 2
Optimized UA-WAAM process parameters.

Arc current (A)	Arc voltage (V)	Travel speed (mm/min)	Wire feed speed (mm/min)	Gas flow rate (L/min)
110	20	200	1800	15

For microstructural analysis, cross-section specimens were sectioned from the WAAM builds using a CNC wire electrical discharge machine. These samples were ground with metallographic abrasive papers and polished with alumina suspensions to achieve a mirror finish. The polished cross-sections were etched with Keller's reagent to reveal the microstructure and pore distribution. The pore size distribution and porosity were evaluated using optical stereomicroscopy on the etched cross-sections. In addition, small samples (approximately 6 mm × 4 mm × 4 mm) oriented in the vertical (build) direction were prepared for electron backscatter diffraction (EBSD). These EBSD specimens were mechanically polished and then electro-polished to remove any surface deformation, enabling accurate grain structure and misorientation measurements.

3. Results

3.1. Effect of ultrasonic frequency on porosity

Porosity is a common defect in wrought and cast aluminum alloys as well as welds, and it is a major source of performance degradation. This holds true for WAAM-processed aluminum alloys, where pores are predominantly formed by hydrogen evolution during solidification [14]. The hydrogen responsible for pore formation primarily originates from contaminants on the substrate or filler wire, with additional contributions possibly from trace moisture in the shielding gas [15].



Fig. 2. Morphology of WAAM walls with (a) excessive welding speed; (b) excessive current; (c) optimized parameters.

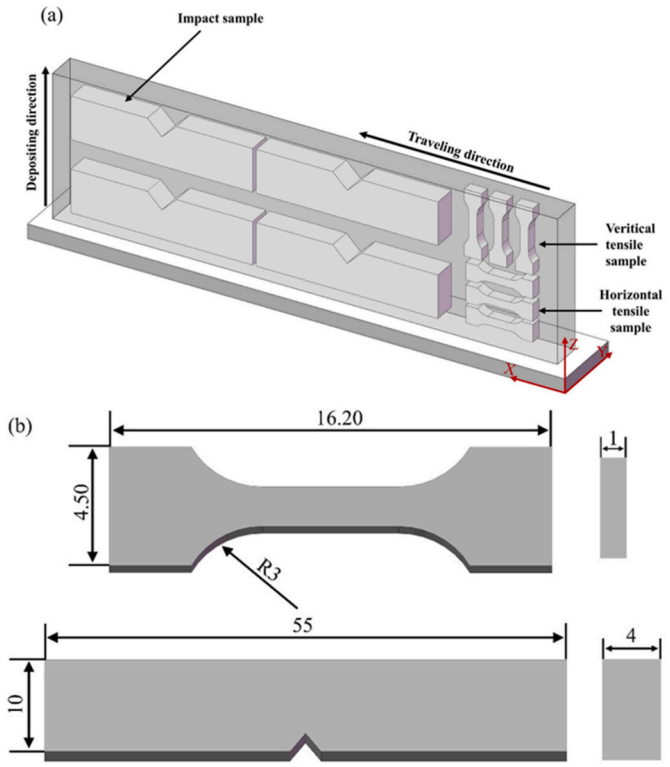


Fig. 3. (a) Sampling locations for the impact and tensile test specimens in the deposited wall; (b) dimensions of the Charpy impact and tensile specimens.

As shown in Fig. 4, introducing ultrasonic vibration during WAAM has a significant effect on pore size, quantity, and distribution (pores appear as white spots). Notably, pores in WAAM builds tend to concentrate at interlayer boundaries—regions that are partially remelted when subsequent layers are deposited. These interlayer zones have higher viscosity than the fully molten pool, which creates unfavorable conditions for dissolved hydrogen to escape. Applying ultrasound greatly influences porosity in these regions. UA-WAAM samples processed at 50 V and 30 kHz exhibit a marked reduction in pore number and size compared to conventional WAAM, underscoring the role of ultrasonic vibration in promoting hydrogen degassing from the melt. Increasing the ultrasonic frequency to 50 kHz (at the same 50 V amplitude) yields a further decrease in porosity, and only a slight additional improvement is observed at 70 kHz—indicating that 50 V, 70 kHz is near the optimal ultrasonic setting for porosity reduction. At the highest tested setting of 50 V and 90 kHZ, the porosity increases again relative to the optimum. This increase occurs because excessive ultrasonic frequency intensifies fluid flow in the molten pool to the point of turbulence; gas bubbles that would otherwise rise out of the pool can become trapped in turbulent vortices and recirculated into the melt, impeding their escape. The sampling area for pore distribution is the stable sedimentary layer section. (The area above the yellow line is the stable sedimentary layer zone).The specific pore density statistics are shown in Table 3.

3.2. Effect of ultrasonic frequency on microstructure

Microstructural examination of the WAAM deposits reveals characteristic patterns in pore distribution and grain morphology. As shown in Fig. 5a, pores in conventional WAAM are primarily concentrated along the interlayer regions (between successive deposited layers). This non-uniform pore distribution has been reported to negatively impact through-thickness (vertical) mechanical properties in WAAM aluminum alloys [16]. For example, Bai et al. [24] observed that in a GTAW-based WAAM 2219 aluminum alloy, tensile ductility in the build direction was significantly reduced due to such interlayer porosity. In the present 5183 alloy, a similar anisotropy is observed: the vertical elongation is lower than the horizontal elongation (as shown later in Section 3.3). Fig. 5b – c shows the as-deposited microstructure of a conventional WAAM sample. The fusion zone of each layer is dominated by continuously grown columnar grains, while a band of equiaxed grains appears at each interlayer region. This results in a periodic alternation of columnar and equiaxed grain regions along the build direction.

To investigate the effects of ultrasound on grain morphology, EBSD analysis was performed on samples produced under various ultrasonic conditions. Fig. 6 presents inverse pole figure (IPF) maps of the grain structures for WAAM and UA-WAAM samples at different ultrasonic settings. In the IPF maps, different colors indicate different crystallographic orientations (uniform color within a region means those grains share the same orientation). Fig. 6a corresponds to the conventional WAAM sample (0 V, 0 kHz), showing a coarse columnar grain structure within each layer and some equiaxed grains at the interlayer boundaries (the black lines delineate successive layer boundaries). Fig. 6b is for a UA-WAAM sample at 50 V, 30 kHz, illustrating that at this relatively low ultrasonic frequency, the large columnar grains within the deposited layers remain largely unrefined—similar to the unmodified WAAM microstructure. Fig. 6c, however, shows that at 50 V, 50 kHz the originally continuous columnar grains are significantly broken up into fine equiaxed grains, indicating effective grain refinement under optimal ultrasonic conditions. Compared to 0 V 0 kHz, grain refinement is more pronounced at 50 V 50 kHz. Grain refinement generally significantly enhances the ability of a material to absorb energy under impact loading. Finer grains result in a higher density of grain boundaries per unit volume, which effectively impede dislocation motion and dissipate more energy. Meanwhile, grain boundaries serve as diffusion pathways for atoms, facilitating the escape of small pores and thereby suppressing crack initiation and rapid propagation. Consequently, grain refinement can improve the impact performance of the material. Fig. 6d and e (50 V, 70 kHz and 50 V, 90 kHz, respectively) demonstrate further evolution of the grain structure with higher ultrasonic intensity: increasingly finer and more equiaxed grains are achieved at 70 kHz, whereas at 90 kHz the grain size remains refined but porosity increases (as noted earlier in Section 3.1).

3.3. Effect of ultrasonic frequency on mechanical properties

The mechanical properties of the deposited alloy were evaluated under both WAAM and UA-WAAM conditions. The results for tensile and impact tests are presented below.

The tensile behavior of WAAM vs. UA-WAAM samples was systematically analyzed. All tensile specimens were extracted from the mid-height of the builds to ensure a consistent microstructure and minimize any sampling variability. Fig. 7a and b shows the engineering stress–strain curves for representative samples built in the vertical and horizontal orientations. The elastic portions of the curves are nearly parallel, indicating that the Young's modulus (~70 GPa for Al) is essentially unchanged by the process conditions. The yield strengths exhibit an orientation dependence: for example, the WAAM samples yield at approximately 120 ± 5 MPa in the vertical orientation and 143 ± 4 MPa in the horizontal orientation. (It should be noted that residual stresses in the as-deposited material approach the yield stress and may influence the apparent yield strength.) Importantly, no significant difference in yield strength is observed between conventional WAAM and UA-WAAM samples beyond the orientation effect.

The ultimate tensile strength (UTS) and elongation (EL) results are summarized in Fig. 7c and d. For the conditions examined, UTS values range from about 230 MPa up to 280 MPa, while elongations range from ~22% up to ~39%. The horizontal specimens generally outperform the vertical ones, consistent with the presence of interlayer porosity that reduces through-thickness ductility. In the horizontal direction,

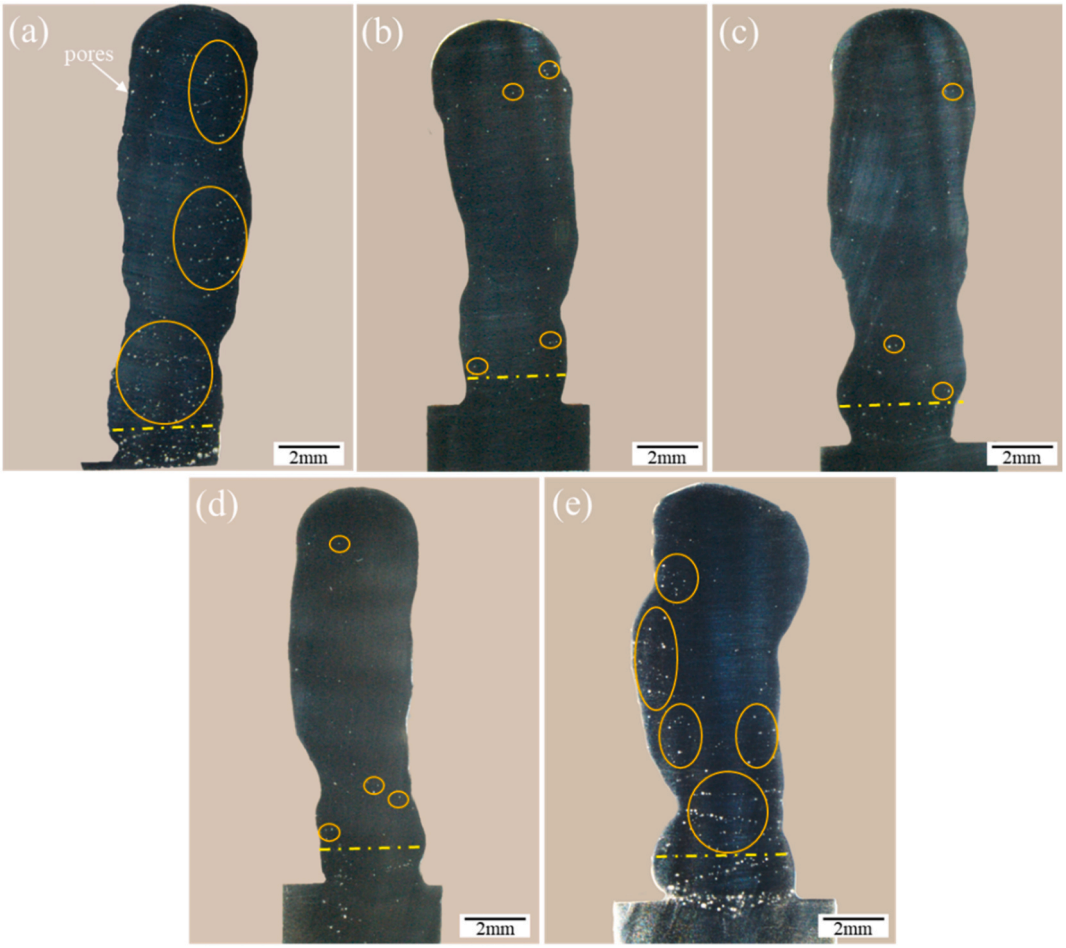


Fig. 4. Pore distributions in the cross-sections of WAAM and UA-WAAM samples under different ultrasonic parameters: (a) 0 V, 0 kHz; (b) 50 V, 30 kHz; (c) 50 V, 50 kHz; (d) 50 V, 70 kHz; (e) 50 V, 90 kHz.

Table 3
Pore density as a function of ultrasonic frequency.

ultrasonic parameters	0v 0 kHz	50v 30 kHz	50v 50 kHz	50v 70 kHz	50v 90 kHz
Number of pores	170	70	60	45	150
Areas/mm^2	110	90	100	95	100
pore density/PPI	1.55	0.78	0.6	0.47	1.5

ultrasonic assistance clearly improves the tensile properties. Conventional WAAM samples achieved an average UTS of ~250.7 MPa with an elongation of ~32%. In comparison, UA-WAAM samples (50 V amplitude) attained UTS values between 262.2 MPa and 279.5 MPa, depending on frequency. The highest strength (279.5 MPa) was achieved at 50 V, 30 kHz. The ductility also improved: the UA-WAAM sample at 50 V, 50 kHz exhibited the greatest elongation of about 39%. Thus, applying ultrasound during WAAM provided up to ~12% higher UTS and ~21% higher EL in the horizontal orientation. In the vertical orientation, the improvements were even more pronounced in percentage terms (as will be summarized in the Conclusions), though the vertical properties remained lower than the horizontal due to the aforementioned anisotropy.

The impact toughness of WAAM and UA-WAAM samples was evaluated using Charpy V-notch tests. All impact specimens were extracted from the mid-thickness of the deposits to maintain consistency in microstructure across samples. Fig. 8 presents the absorbed impact energy (notch toughness) for samples fabricated under various ultrasonic parameters. It is evident that ultrasonic excitation substantially increases the impact toughness. As the ultrasonic frequency increases (at a fixed 50 V amplitude), the impact toughness of the alloy rises steadily, reaching a maximum at the 50 V, 70 kHz condition. Beyond this point, a slight decline is observed at 90 kHz, although the toughness at 50 V, 90k Hz remains about 21.7% higher than that of the non-ultrasonicated WAAM material.

Quantitatively, the conventional WAAM sample exhibits an impact toughness of ~206 J/cm^2. Introducing ultrasonic vibration yields significantly higher values: at 50 V, 30 Hz, the toughness increases to ~323 J/cm^2; at 50 V, 50 Hz, it further rises to ~358 J/cm^2. The peak toughness of 363 J/cm^2 is achieved at 50 V, 70 kHz, corresponding to about a 76% improvement over the WAAM baseline. At the highest ultrasonic setting (50 V, 90 kHz), the toughness drops to ~239 J/cm^2, which, while lower than the optimum, is still above the value for conventional WAAM. These results indicate that an intermediate ultrasonic frequency provides the greatest benefit to impact toughness, whereas too high of a frequency can be detrimental, consistent with the porosity trend noted earlier.

4. Discussion

Ultrasonic vibration during welding introduces two primary physical phenomena in the molten metal: cavitation (the formation, growth, and violent collapse of gas/vapor bubbles) and acoustic streaming (sound-induced fluid flow) [17]. Cavitation occurs when high-intensity ultrasound causes vapor bubbles to rapidly nucleate and implode in the liquid [18]. Acoustic streaming is the steady flow generated by the attenuation of acoustic energy in the liquid and the resulting pressure gradients [19].

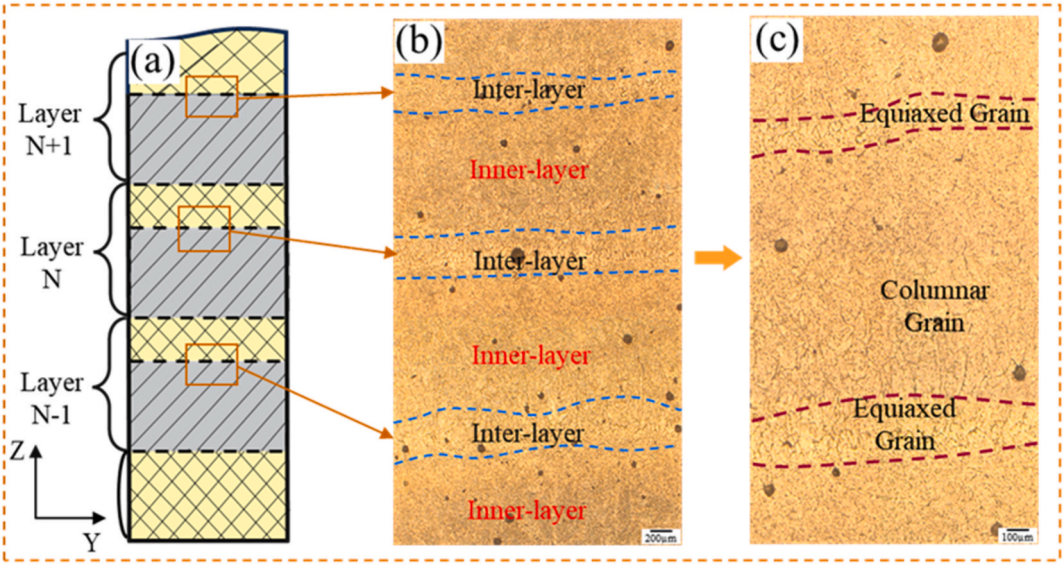


Fig. 5. (a) Schematic illustration of the metallographic structure distribution in a WAAM deposit; (b) microstructure of conventional WAAM (no ultrasound) at 50 × magnification; (c) microstructure of conventional WAAM at 100 × magnification (showing columnar grains within layers and equiaxed grains at an inter-layer boundary).

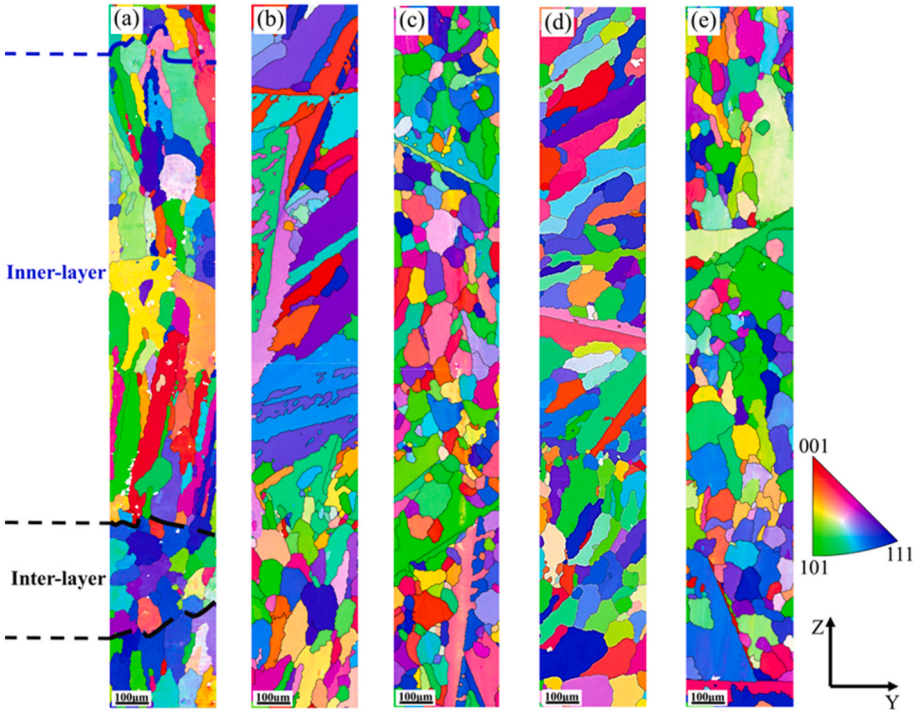


Fig. 6. Inverse pole figure (IPF) maps of grains in WAAM and UA-WAAM samples under different ultrasonic conditions: (a) 0 V, 0 kHz; (b) 50 V, 30 kHz; (c) 50 V, 50 kHz; (d) 50 V, 70 kHz; (e) 50 V, 90 kHz.

The intensity of cavitation in the weld pool depends on the acoustic pressure amplitude exceeding a threshold; generally, higher acoustic pressure produces stronger cavitation, though at very high amplitudes the effect can saturate or become nonlinear due to phenomena like acoustic shielding by bubbles [20].

The acoustic pressure amplitude P required to trigger cavitation in an aluminum alloy melt can be estimated by the following relation [21]:

$$P_a = P_0 - P_v + \frac{2\sigma}{R_0},$$

where P_0 is the hydrostatic pressure (101 kPa), P_v is the saturation vapor pressure of the liquid aluminum (~6.8 kPa at the melting point), σ is the surface tension (0.91 N/m for Al), and R_0 is the initial bubble nucleus radius (on the order of 10 μm). Once this cavitation threshold is reached, vapor bubbles form explosively and then collapse, generating intense localized shock waves in the molten pool [22,23]. During cavitation, the rapid expansion and collapse of bubbles produce micro-jets and pressure pulses that vigorously agitate the molten metal [24]. As illustrated in Fig. 9, these effects serve dual purposes: (a) the agitation breaks up larger pores into smaller ones and distributes them more uniformly, and (b) the micro-jetting action helps expel bubbles from the liquid before

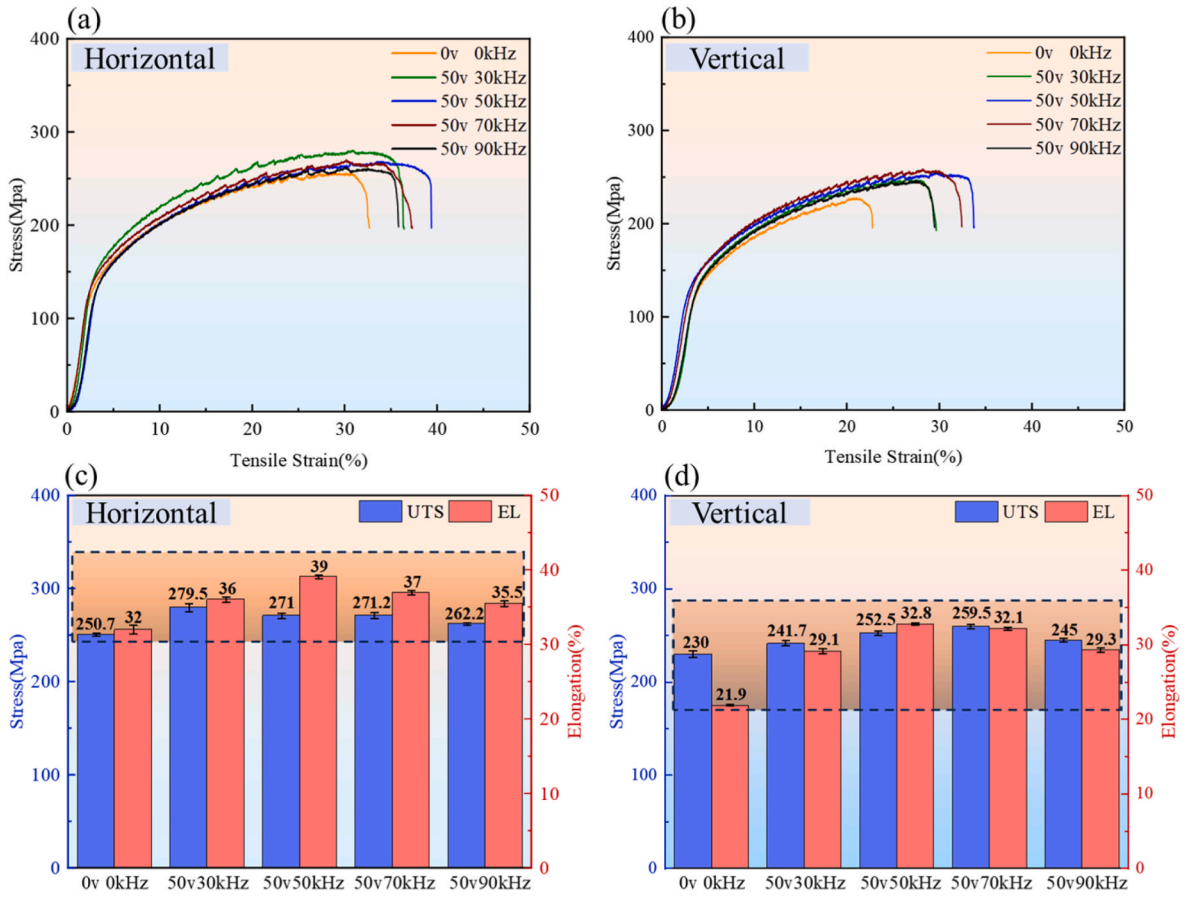


Fig. 7. Tensile properties of WAAM and UA-WAAM 5183 aluminum alloy samples: (a, b) engineering stress–strain curves for vertical and horizontal specimens; (c, d) average ultimate tensile strength and elongation (percent) for each condition (error bars represent ±1 SD).

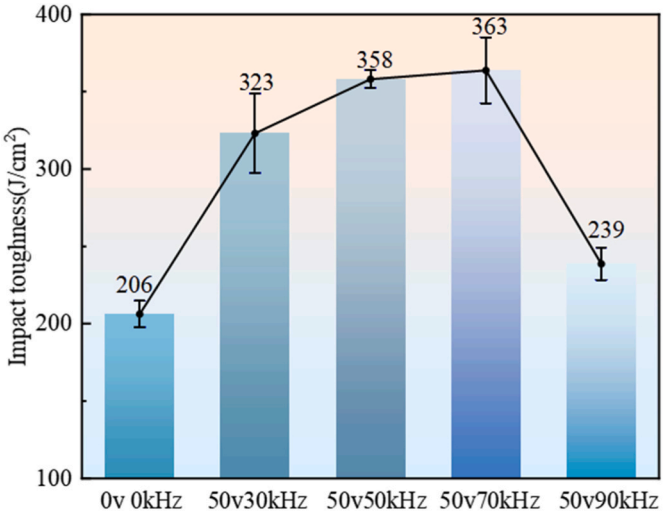


Fig. 8. Impact toughness of WAAM and UA-WAAM samples as a function of ultrasonic excitation frequency (0 kHz = conventional WAAM, higher frequencies at 50 V amplitude).

they can be trapped by solidification [24,25]. The introduction of ultrasound leads to higher acoustic pressure within the melt pool at higher frequencies, resulting in intensified transient cavitation and an increased population of fine bubbles. When the acoustic pressure reaches an equilibrium state, it significantly promotes bubble growth. Since these bubbles do not collapse, they continue to grow under the stable acoustic pressure. Consequently, the vibration frequency generated by ultrasound induces intense agitation within the melt, enabling the bubbles to escape rapidly. The condition of 0 V and 0 kHz, where no ultrasound was applied, reveals the presence of numerous large pores in the as-built structure (as indicated in Fig. 4). From 50 V to 30 kHz to 50 V and 90 kHz, the large bubbles essentially disappear, and fine bubbles emerge. However, at 50 V and 90 kHz, the cavitation threshold is reached, where the excessively high vibration frequency prevents many fine bubbles from escaping the melt pool. In contrast, at 50 V and 70 kHz, the stable acoustic pressure and the 70 kHz vibration frequency facilitate the rapid escape of bubbles.

Acoustic streaming further contributes to these effects. The dissipation of ultrasonic energy in the melt drives a steady convective flow that significantly increases fluid motion in the weld pool [26]. As shown in Fig. 9a, this streaming provides an additional force to sweep out bubbles and pores from the molten pool. In the absence of ultrasound, fluid flow in aluminum weld pools is largely limited to Marangoni convection, which is relatively weak and often insufficient to remove pores before solidification [27]. Consequently, conventional WAAM deposits tend to exhibit higher porosity. Under ultrasonic excitation, the combined action of cavitation-induced shock waves and acoustic streaming is the primary mechanism behind the observed reduction in pore size and number in UA-WAAM samples.

The microstructure and fracture behavior of the alloy are also markedly influenced by ultrasonic assistance. To elucidate the fracture mechanisms, SEM fractography was performed on the broken Charpy specimens. Fig. 10 compares the fracture surfaces of a conventional WAAM sample and UA-WAAM samples at various ultrasonic frequencies. Fig. 10a shows the WAAM impact fracture surface, which contains numerous pores scattered throughout. In contrast, the UA-

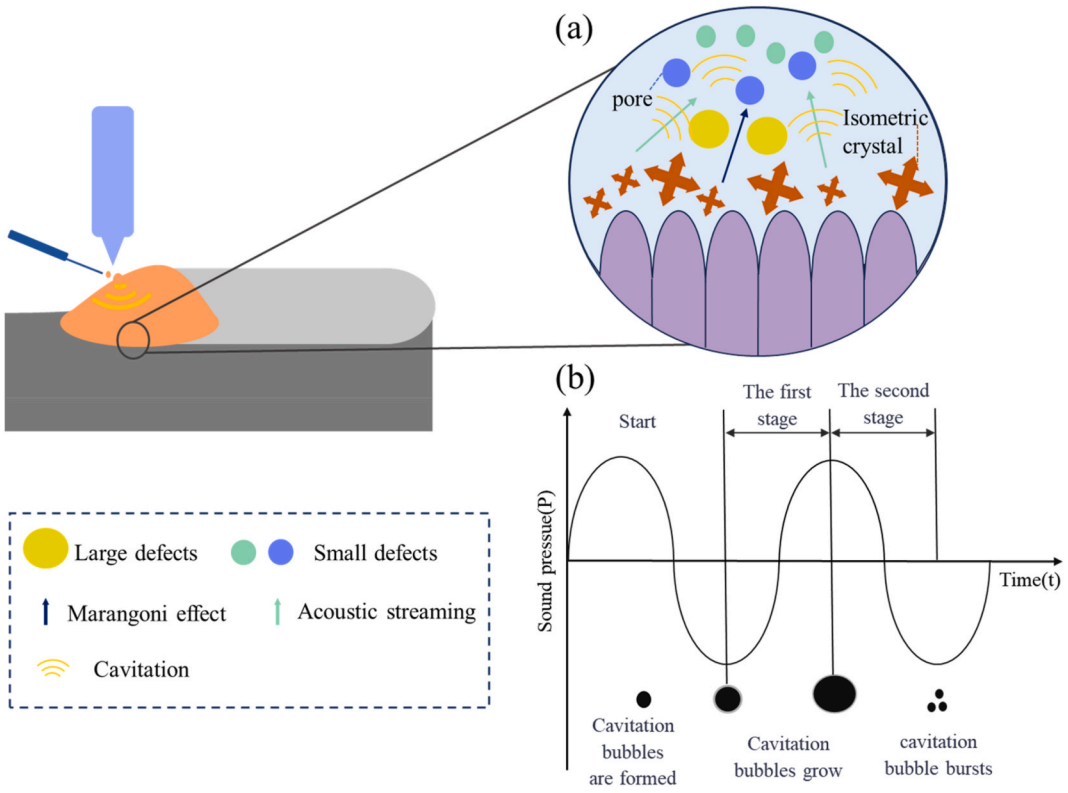


Fig. 9. (a) Schematic illustration of pore evolution in the molten pool(cavitation and acoustic streaming promote bubble fragmentation and removal); (b) Schematic illustration of bubble cavitation evolution.

WAAM fracture surfaces (Fig. 10b, c, d, e corresponding to 30 kHz, 50 kHz, 70 kHz, 90 kHz, respectively, all at 50 V) exhibit progressively fewer and smaller pores as the ultrasonic frequency increases. This indicates that higher ultrasonic intensity produces a cleaner fracture surface with fewer internal voids, consistent with the porosity measurements. All samples display a ductile fracture morphology characterized by dimples (micro-void coalescence) on the fracture surface. However, the dimple size and distribution differ between WAAM and UA-WAAM. Quantitative analysis of dimple size distribution (determined by the average dimple diameter) on impact fracture surfaces was performed under various ultrasonic parameters. For each parameter set, the same scanning area within the radial zone of the fracture surface was selected for dimple size evaluation. The average dimple diameters obtained are as follows: 11 μm at 0 V/0 kHz, 10 μm at 50 V/30 kHz, 7 μm at 50 V/50 kHz, 7 μm at 50 V/70 kHz, and 8 μm at 50 V/90 kHz. The WAAM sample shows larger, more irregular dimples interspersed with voids, whereas the UA-WAAM samples show a higher density of finer, deeper dimples. The refined dimple structure in the ultrasonically treated samples is indicative of increased energy absorption during fracture, correlating with their improved impact toughness. In other words, the elimination of large defects (pores), the refinement of the pore structure, and the relaxation of dislocation density in UA-WAAM collectively contribute to a tougher and more crack-resistant fracture behavior. On the other hand, the static tensile strength did not increase as dramatically with ultrasonication because, although porosity was reduced, the remaining small pores were uniformly dispersed and did not significantly impede the tensile load-bearing cross-section.

EBSD-based kernel average misorientation (KAM) analysis was used to assess the residual plastic strain and dislocation density in the WAAM vs. UA-WAAM deposits. Fig. 11 shows KAM maps for the WAAM sample and several UA-WAAM samples. In the WAAM map (Fig. 11a), the majority of the area is colored green with some red patches, indicating a relatively high density of geometrically necessary dislocations (GNDs) and localized residual stress regions. In contrast, the UA-WAAM maps (Fig. 11b–e, corresponding to 50 V at 30 kHz, 50 kHz, 70 kHz, 90 kHz) have far fewer red/green regions; most of the area is blue, indicating generally lower misorientation angles and thus lower dislocation density. The areal fraction of high-misorientation regions decreases with increasing ultrasonic frequency up to 70 kHz, suggesting that ultrasonic vibration effectively relieves residual stresses and reduces dislocation content. This observation is consistent with independent neutron diffraction measurements of residual stress (not shown). The average KAM value for the conventional WAAM sample was measured to be ~0.53. By comparison, the UA-WAAM samples at 50 V showed average KAM values of approximately 0.36 (30 kHz), 0.43 (50 kHz), 0.32 (70 kHz), and 0.42 (90 kHz). A higher dislocation density generally leads to increased yield strength and elongation. Although the sample processed without ultrasonic treatment (0 V, 0 kHz) exhibits the highest dislocation density compared to those under other ultrasonic parameters and would therefore be expected to show the highest yield strength and elongation, its concomitant high pore density ultimately results in a reduction of these mechanical properties (as shown in Fig. 7). The lowest dislocation density was achieved at the 50 V, 70 kHz condition (KAM ~0.32), which is also the condition that produced the highest impact toughness. Interestingly, the 50 V, 50 kHz sample had a higher KAM (more dislocations, ~0.43) than the 50 V, 30 kHz sample (~0.36), yet 50 kHz yielded better toughness because it had significantly lower porosity. This suggests that porosity is the dominant factor governing impact toughness of these additively manufactured samples, while dislocation density (residual stress) plays a secondary strengthening role.The contribution of dislocations to the alloy's strength can be described by a Taylor hardening relationship [28,29]:

$$\Delta\sigma = M\,\alpha\,G\,b\,\sqrt{\rho},$$

where σ is the increase in yield strength due to dislocations, M is the Taylor factor, α is a dimensionless constant (of order 0.3–0.5 for

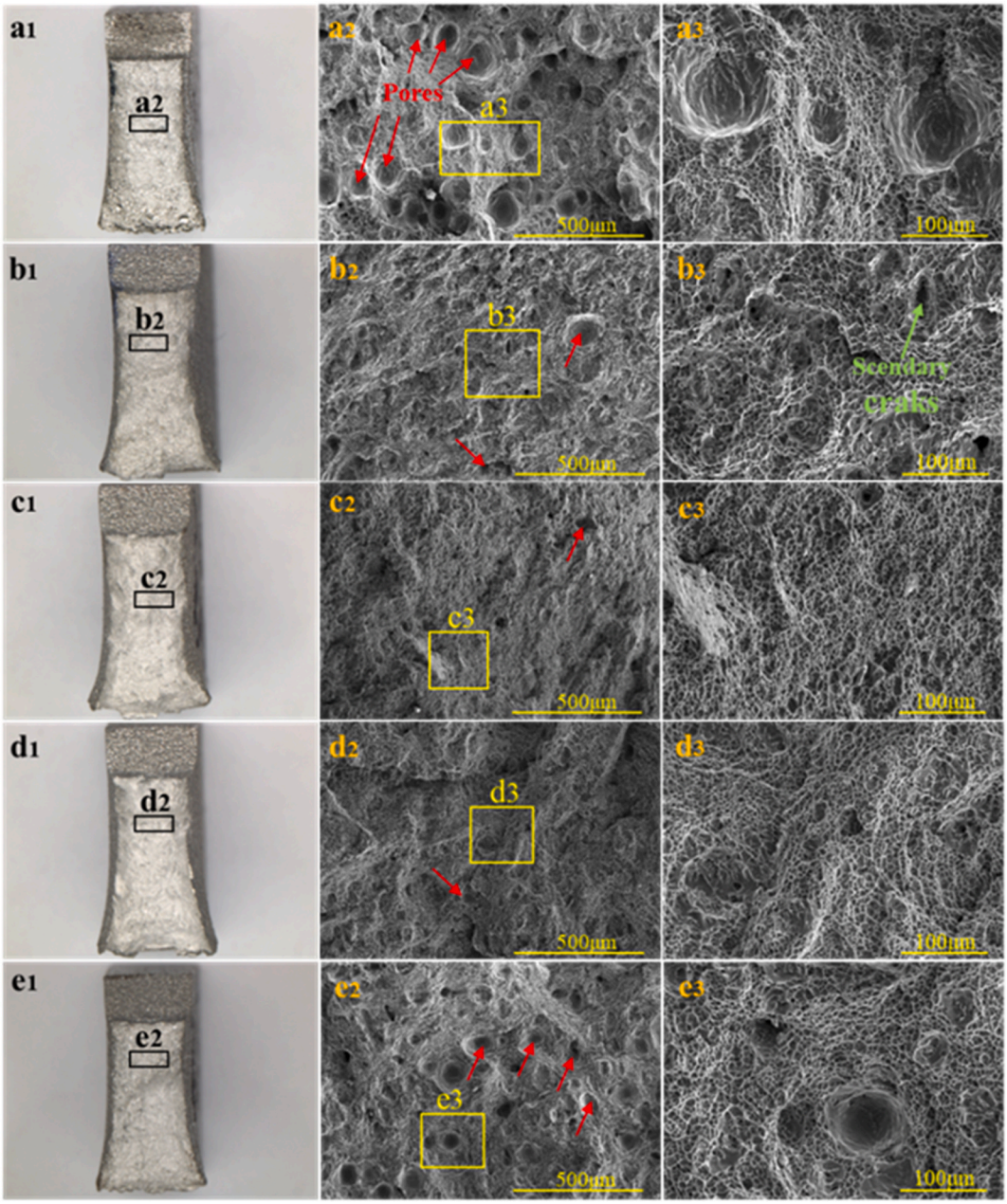


Fig. 10. Macroscopic and SEM fractography of impact fracture surfaces for WAAM and UA-WAAM samples: (a1-a3) WAAM (0 V, 0 kHz); (b1-b3) UA-WAAM (50 V, 30 kHz); (c1-c3) UA-WAAM (50 V, 50 kHz); (d1-d3) UA-WAAM (50 V, 70 kHz); (e1-e3) UA-WAAM (50 V, 90 kHz).

aluminum alloys [8]), G is the shear modulus of aluminum, b is the Burgers vector, and ρ is the dislocation density. According to this relationship, a lower dislocation density (lower ρ) corresponds to lower dislocation strengthening, which can translate to higher ductility and toughness. In our observations, UA-WAAM samples have lower dislocation densities (especially at the optimal 70 kHz condition), which helps explain their higher toughness. In essence, ultrasonic vibration during WAAM promotes the formation of sub-grain cellular structures that reorganize dislocations into walls and cell boundaries, thereby relieving internal stresses. These microstructural substructures (low-misorientation subgrains) reduce the overall dislocation density and residual stress in the deposits [30,31]. Together with the reduction in porosity and grain refinement, this leads to the improved mechanical performance (particularly the impact toughness) observed in the UA-WAAM material.

5. Conclusions

(1) Grain refinement and residual stress mitigation: Ultrasonic vibration during WAAM refines the grain structure by breaking up columnar grains into finer equiaxed grains. EBSD analysis showed that UA-WAAM samples have lower kernel misorientation and thus lower dislocation (residual stress) densities than WAAM samples. This indicates that ultrasonic assistance alleviates residual stresses in the deposits by promoting the formation of dislocation cell structures and subgrain boundaries.

(2) Enhanced tensile properties: UA-WAAM samples exhibit improved tensile performance compared to WAAM. In the vertical (build) direction, the ultimate tensile strength and elongation of UA-WAAM were increased by approximately 12% and 49%, respectively. In the horizontal direction, both UTS and elongation were improved by about 21%. These enhancements are attributed

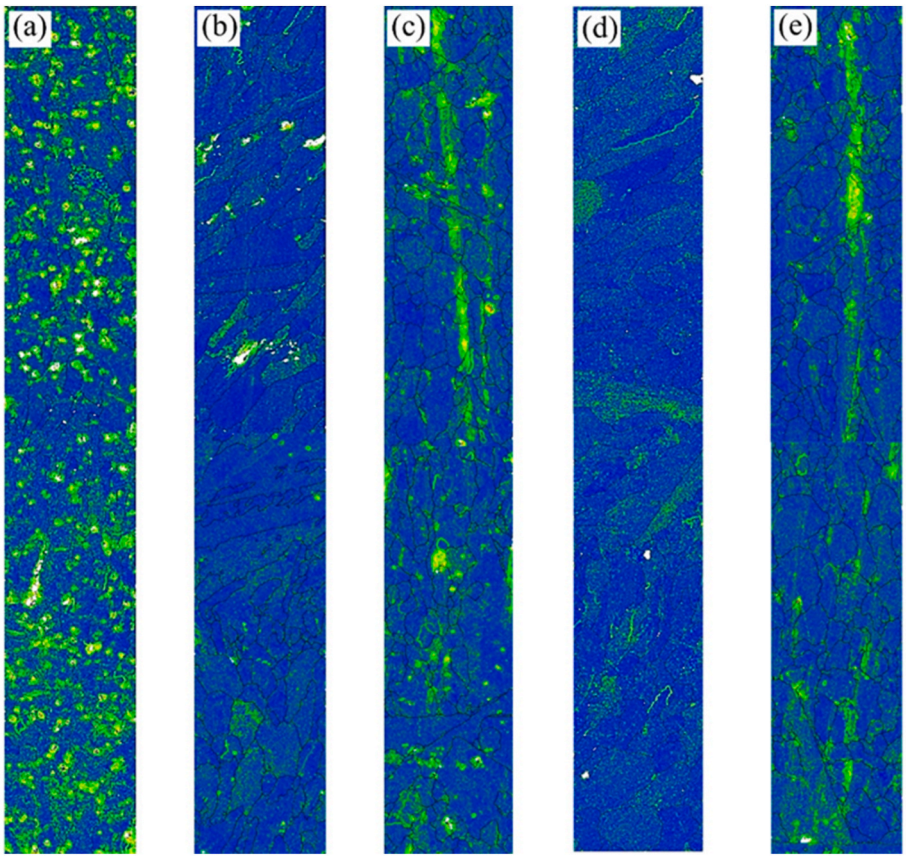


Fig. 11. EBSD kernel average misorientation (KAM) maps of WAAM and UA-WAAM samples under various processing conditions, illustrating residual strain distribution: (a) WAAM (0 V, 0 Hz); (b) UA-WAAM (50 V, 30 kHz); (c) UA-WAAM (50 V, 50 kHz); (d) UA-WAAM (50 V, 70 kHz); (e) UA-WAAM (50 V, 90 kHz).

to the refined microstructure and reduced defect content in the ultrasonically assisted builds.

(3) Significantly improved impact toughness: The Charpy impact toughness of the 5183 alloy was dramatically increased (by up to 76%) under UA-WAAM processing. The impact energy absorbed climbed steadily with increasing ultrasonic frequency and peaked at the intermediate frequency (50 V, 70 kHz) before slightly declining at the highest frequency tested. Fractography confirmed ductile failure modes for all samples, with UA-WAAM specimens showing finer and deeper dimples and far fewer voids on the fracture surface. The greatly improved impact toughness in UA-WAAM is primarily attributed to the marked reduction in porosity and the resulting increase in crack propagation resistance.

Declaration of competing interest

The authors declared that they have **No conflicts of interest** to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted entitled "Ultrasonic-Assisted Wire Arc Additive Manufacturing Significantly Improves Impact Toughness of 5183 Aluminum Alloy".

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